S13 Wave Motion | 2016 Q6

Although most candidates applied the correct formula to find · in (a)(i), many made mistakes in getting the fringe separation from the pattern given or in converting the units. In (a)(ii), quite a number of the candidates confused slit width with slit separation. Not many pointed out explicitly that the diffracted waves from the slits have to overlap for interference to occur. Some candidates wrongly applied ·y = ·D/a to find the separation between bright spots in (b)(i). Candidates' performance in (b)(ii) was poor. The patterns drawn were either with uniform separation between bright spots, not symmetrical about the centre or without the central bright spot.

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This question tested candidates' basic knowledge of wave motion. Candidates' performance was satisfactory. Part (a) was well answered. In (b)(i), some candidates only stated 'crest meets trough at Q' to explain why a minimum occurred there. Quite a number of them did not know that in (b)(iii) the amplitude of the water waves would decrease gradually with the distance from the source.

(No Section B candidate-performance note.)